Pediatrics

Growth & Immunization

BY

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INDEX

Growth and its Disorders

Bone Age Estimation

Developmental stages

- Tanner and Whitehouse described 8 to 9 stages of development of ossification centres.

- **'Maturity scoring' assigned as follows:**

    - 50% for carpal bones
    - 20% for radius and ulna
    - 30% for phalanges.

Age specific Radiographs

- **Birth:** Radiograph of Knee

- **3-9 months:** Radiograph of shoulder is most helpful.

- **1-13 years:** A single film of hands and wrists

- **12- 14 years:** Radiographs of elbow and hip give helpful clues.

Eruption of teeth

Primary dentition (Deciduous Teeth)			
Tooth	**Upper Jaw Eruption(Months)**	**Lower Jaw Eruption (Months)**	**Time of Fall (Years)**
Central incisors	8-12	6-10	6-7
Lateral incisors	9-13	10-16	7-8
First molar	13-19	14-18	9-11
Canine	16-22	17-23	9-12
Second molar	25-33	23-31	10-12

Permanent dentition		
Tooth	**Upper Jaw Eruption(Years)**	**Lower Jaw Eruption (Years)**
First molar	6-7	6-7
Central incisors	7-8	6-7
Lateral incisors	8-9	7-8
Canine	11-12	10-12
First premolar	10-11	10-12
Second premolar	10-12	10-12
Second molar	12-13	11-13
Third molar	17-21	17-21

Approximate anthropometric values by age			
Age	**Weight (kg)**	**Length or height (cm)**	**Head circumference (cm)**
Birth	3	50	34
6 months	6 (doubles)	65	43
1 year	9 (triples)	75	46
2 years	12 (quadruples)	87	48
3 years	15	95	49
4 years	16	100	50

Situations indicating a problem or suggest risk on growth curve:

- A child's growth curve crosses a major z-score line or two major centile lines.
- A sharp incline or decline in the child's growth curve.
- A growth curve that remains flat (stagnant)

Disorders of Growth

Short Stature:

Definition: Short stature is defined as height below the third centile or more than two standard deviation scores (SDS) below the median height for age and gender (<-2SDS) according to the population standard.

Causes:

- Cerebral palsy
- Cystic fibrosis, asthma
- Malabsorption
- Hypothyroidism
- Cushing syndrome
- Pseudohypoparathyroidism
- Precocious or delayed puberty
- Rickets
- Psychosocial dwarfism
- Skeletal dysplasia
- Genetic syndromes

Assessment of body proportion:

- The proportionality is assessed by the upper segment (US): lower segment (LS) ratio.

- **Normally, the US:LS ratio is:**

- 1.7 at birth
- 1.3 at 3 years
- 1.1 by 6 years
- One by 10 years
- 0.9 in adults

- **Increase in the US:LS ratio** is seen in rickets, achondroplasia, and untreated congenital hypothyroidism.

- **Decrease in the US:LS ratio** is seen in spondyloepiphyseal dysplasia and vertebral anomalies.

Specific Etiologies:

Familial short stature: The child is short as per definition (height <3rd centile) but is normal according to his genetic potential determined by the parent's height.

Constitutional growth delay: These children are born with a normal length and weight and grow normally during the first year.

Their growth then decelerates so that the height and weight gradually fall below the 3rd centile.

The onset of puberty and adolescent growth spurt is delayed in these children, but the final height is within normal limits.

Feature	Constitutional growth delay	Familial short stature
Height	Short	Short
Height velocity	Normal	Normal
Family history	Delayed puberty	Short stature
Bone age	Less than chronological age	Normal
Puberty	Delayed	Normal
Final height	Normal	Low but normal for target height

Tall Stature:

Definition: Height above two standard deviations for age and gender according to the population-specific growth reference is considered as tall stature.

Causes:

- Familial tall stature
- Obesity
- Klinefelter syndrome, Marfan syndrome, homocystinuria
- Growth hormone excess, hyperthyroidism
- Precocious puberty
- Sex steroid deficiency

Immunization

Types of Vaccine	
Description	**Example**
Live attenuated	BCG, oral typhoid (Ty21a), OPV, measles, MMR, varicella, rotavirus, yellow fever, influenza (intranasal), hepatitis A
Killed or inactivated	Whole cell pertussis, IPV, rabies, hepatitis A, influenza, SARS-CoV-2
Toxoid	Diphtheria, tetanus
Capsular polysaccharide	Typhoid (Vi), Hib, meningococcal, pneumococcal
Conjugated	Hemophilus influenzae b, meningococcal, pneumococcal
Recombinant	Acellular pertussis, hepatitis B, papillomavirus
Viral vector borne	SARS-CoV-2

Vaccination Route

- Locations for intramuscular (IM) administration include anterolateral aspect of thigh in infants and deltoid in upper arm in older children and adults.

- Gluteal administration is avoided to reduce risk of sciatic nerve injury.

- Hepatitis B is particularly not given in gluteal region as this has reduced efficacy.

- Two vaccines may be given in same thigh but separated by 1 inch.

- Separate sites are used when administering a vaccine and immunoglobulin.

Principles of Immunization

- Different live (oral, parenteral, intranasal) vaccines may be given simultaneously, or 4 weeks apart.

- Different types of inactivated or subunit vaccines may be administered simultaneously or at any interval between their doses.

- A minimum interval of 4 weeks between two doses of the same vaccine is necessary to ensure adequate immune responses. An exception is the rabies vaccine.

- Live-and an inactivated vaccine can be administered simultaneously or at any interval of time.

- Patients should be observed for allergic reactions for 15-20 minutes after receiving immunization.
- Immunization is not contraindicated in minor illness, prematurity, allergies, malnutrition or during infection and antibiotic therapy.

Mission Indradhanush

The mission derives its name from the seven diseases prevented through these vaccines (BCG, OPV, pentavalent and TT vaccines) and focuses on complete immunisation of all children less than 2 years old and in pregnant women.

Vaccines covered

- BCG
- OPV
- Pentavalent (Includes Diphtheria, Pertussis, tetanus, Hib, Hepatitis B)
- TT
- Measles
- Rubella
- Rotavirus diarrhea
- Pneumococcal conjugate
- JE

Age	National Immunization Program
At Birth	BCG, OPV-0, HBV-0
6 weeks	OPV-1, Pentavalent-1, Rota-1, fIPV-1, PCV-1
10 weeks	OPV-2, Pentavalent-2, Rota-2
14 weeks	OPV-3, Pentavalent-3, Rota-3, fIPV-2, PCV-2
9 months	MR-1, JE-1, PCV-3, fIPV-3
16-18 months	DTwP-Booster 1, OPV-Booster, JE-2, MR-2
5-6 yr	DTwP-Booster 2
10-12 yr	Td
9-14 yr	HPV-1, HPV-2
16 yr	Td

BCG VACCINE

- **Composition:** Live-attenuated M. bovis Calmette-Guérin (Danish 1331 strain in India).
- **Administration:** 0.1 mL, Intradermal at left shoulder, forming a wheal.
- **Immunity:** Primarily cell-mediated; maternal antibodies do not interfere.
- **Storage:** 2-8°C; rapid potency loss after reconstitution.

- **Contraindication:** Cellular immunodeficiency, symptomatic HIV

- **Preparation and dosage:** Reconstituted with sterile normal saline, each dose (0.1 mL) contains 0.1–0.4 million live viable bacilli.

Immunological Effects and Efficacy:

- **Induced Immunity:** Primarily induces cell-mediated immunity.

- **Efficacy:**

    - **Primary infection:** Low protective efficacy (~40%)

    - **Pulmonary infection:** Variable efficacy (8–79%)

    - **Overall forms of tuberculosis:** (~50%)

    - **Severe forms of tuberculosis** (miliary tuberculosis, tubercular meningitis): Over 50% protection and reduces the risk of mortality.

- Since maternal antibodies do not interfere with cellular immune responses, BCG is given at birth ensuring compliance and early protection.

- Conventionally, BCG vaccine is administered at insertion of the deltoid on the left shoulder to allow easy identification of its scar.

- Intradermal injection using a 26 G needle raises a 5–7 mm wheal.

- Bacilli multiply to form a papule by one week that enlarges to 4–8 mm, ulcerates by 5–6 weeks and heals by scarring by 6–12 weeks.

- Inadvertent subcutaneous injection causes persistent ulceration and ipsilateral axillary or cervical lymphadenopathy.

- Children with severe immunodeficiency may develop disseminated BCG disease 6–12 months after vaccination.

Poliomyelitis Vaccine

Oral Polio Vaccine (OPV)

- **Live attenuated** oral poliovirus vaccine (OPV) developed by **Sabin**.
- Contains **magnesium chloride** as a **stabilizing agent**.
- OPV vaccine is given as two drops oral at birth or < 2 weeks (zero dose); 3 doses at **6, 10 and 14 weeks (primary); one booster dose at 15-18 months.**

- Breastfeeding and mild diarrhoea are not contraindicated for OPV administration.
- Children with immunodeficiency and pregnant women should not receive the vaccine.

- Serotype 2 is the most immunogenic of the three serotypes, and interferes with the immunogenicity of serotypes 1 and 3.

- VAPP is observed chiefly due to OPV type 3 (in recipients) and OPV type 2 (in contacts)

- **Seroconversion rates** after 3 doses of OPV are **highest for serotype 2** (90%) and lowest for serotype 3 (70%).

Fractional Inactivated Polio Vaccine (fIPV)

- Intradermal fractional Inactivated Polio Vaccine (fIPV) is used as a dose-sparing strategy in immunization programs, especially in countries with limited vaccine supply.

- The fractional dose (fIPV) is administered intradermally (usually 0.1 mL) and 0.5 mL intramuscular or subcutaneous.

- Three intradermal fractionated doses at 6 weeks, 14 weeks and 9-12 months.

- The standard intramuscular (I/M) IPV dose is 0.5 mL. So, 0.1 mL (fIPV) is one-fifth (1/5th) of the 0.5 mL (I/M) dose.

DPT Vaccine

- Diphtheria vaccine contains diphtheria toxin (DT) inactivated by formalin and adsorbed on aluminium hydroxide, the adjuvant.

- Quantity of toxoid in vaccine, expressed as limit of flocculation (Lf) content is 20-30 Lf of DT, 5-25 Lf of tetanus toxoid (TT) and >4 IU of whole cel killed pertussis.

- **Composition:** Diphtheria toxoid (DT), Tetanus toxoid (TT), and either: Whole-cell pertussis (DTwP) or Acellular pertussis (DTaP)
- **Dose & route:** 0.5 mL; intramuscular at anterolateral aspect of mid-thigh.

- **Schedule:** DTwP at 6, 10, 14 weeks; Boosters at 15-18 months and 5 years.

- **Preference:** DTwP is preferred for primary immunization due to better immune response.

- **Contraindication:** anaphylaxis after previous dose, encephalopathy within 7 days of previous dose.

Acellular Pertussis Vaccine (DTaP)

- Since active pertussis toxin and endotoxin are responsible for the high incidence of adverse events associated with DTwP, various types of purified acellular pertussis vaccines have been developed, termed DTaP.

- The available DTaP vaccines contains atleast 3 pathogenic antigens, commonly **pertussis toxin (PT)** and one or more additional pertussis antigens, like **filamentous hemagglutinin (FHA), pertactin, fimbrial protein and a nonfimbrial protein.**

- Each dose of the vaccine contains at least 4 IU (10-25 mg) of PT component and 6.7-25 Lf of DT.

- DTaP vaccine is not recommended as part of the National Program in India due to its cost.

- **Contraindication for DTaP are the same as for DTwP**; and the vaccine should not be administered if a previous dose of DTwP or DTaP was associated with immediate anaphylaxis, or the development of encephalopathy within 7 days of vaccination.

- These children should **complete immunization with DT** instead of DTwP or DTaP.

MMR Vaccine

- **Composition:** Live-attenuated measles, mumps, and rubella viruses.

- **Mechanism:** Induces mucosal, systemic humoral, and cellular immunity.

- **Dose, route:** 0.5 mL; subcutaneous at right upper arm or anterolateral thigh.

- **Schedule:** Two doses—9-12 months and 15-24 months.

- **Storage:** Lyophilised; 2-8^{0}C use within 4-6 hours of reconstitution.

Hepatitis B Vaccine

- **Composition:** Contains HBsAg produced by recombinant DNA technology in yeast, adsorbed on aluminium salt.

- **Dose:** 0.5 mL (10 μg); 1 mL for immunosuppressed or high-risk patients.

- **Route:** Intramuscular.

- **Site:** Anterolateral thigh or deltoid; avoid gluteal region.

- **Schedule:**
  - **National:** Birth, 6 weeks, 10 weeks, 14 weeks.
  - **Catch-up:** 0, 1, and 6 months.

- **Contraindications:** Anaphylaxis after previous dose.

- **Storage:** 2-8°C; do not freeze.

Mother is known HBsAg negative	Vaccination of the child at 6 weeks.
Mother's status is not known	Vaccinate the newborn within a few hours of birth.
Mother is known HBsAg positive	Vaccinate within few hours of birth , along with hepatitis B immunoglobulin (HBIG) within 24 hr of birth

Rotavirus Vaccine

- Rotavirus is the primary cause of diarrhoea in **infants and toddlers**, accounting for 6-45% of **diarrhoea-related hospitalisations** in Indian children.

- Natural infection does not protect against reinfection or severe disease.

Vaccine Types

Box 10.10: Rotavirus Vaccines

Vaccine name	Rotavac	RotaTeq	Rotarix	Rotasiil
Name by type	Indian neonatal (116E); human bovine monovalent	Human bovine pentavalent (RV5)	Human monovalent (RV1)	Bovine pentavalent; thermostable
Number of strains	1	5	1	5
Schedule[1]				
National program	3 doses at 6, 10 and 14 weeks	Not included	Not included	Not included
IAP 2021	3 doses at 6, 10 and 14 weeks	3 doses at 6, 10 and 14 weeks	2 doses at 6 and 10, or 10 and 14, weeks[2]	3 doses at 6, 10 and 14 weeks
Dose, route	5 drops, oral	2 mL (liquid), oral	1 mL (lyophilized); 1.5 mL (liquid), oral	2.5 mL, oral
Catch up	National program: Up to 1 year of age			

Adverse Reactions

- **Common:** Fever, diarrhoea, vomiting.
- **Rare:** Intussusception.

Contraindications

- Past history of intussusception.
- Severe immunodeficiency.

Precautions

- Avoid vaccination during ongoing diarrhoea or moderate illness

Storage

- **Store at:** 2-8°C.
- **Protect from:** Heat and light.
- **Use within**: 4 hours of reconstitution or opening.

Human Papillomavirus (HPV) Vaccine

- Vaccines contain self-assembling virus-like particles (VLP) with recombinant Ll, the major capsid protein.

- VLP based vaccines protect against 90% new infections with the included serotypes.

- The vaccines do not protect against serotypes with which infection has already occurred before vaccination.

- Serotype 16 and 18 cause 80-85% of invasive cervical cancers.

- **Non oncogenic serotype:** Serotype 6 and 11 cause about 90% of anogenital warts.

- Two vaccines are currently licensed:

  - **Gardasil (HPV 4)** is a **quadrivalent vaccine** active against HPV strains 6, 11, 16 and 18 while

  - **Cervarix (HPV2)** is a bivalent vaccine targeting only HPV 16 and 18.

- **Gardasil** prevents serotype-related anogenital warts, vulval and vaginal cancers and vulvar intraepithelial neoplasia.

- Gardasil 9 additionally protects against anal, oropharyngeal and head and neck cancers caused by HPV.

- Both vaccines are highly immunogenic and persistent protection **for atleast 8-10 yr**

- The recommended age for initiation of vaccination is 11-13 yr, with catch up vaccination permitted up to 45 yr of age .

- Both vaccines are contraindicated in patients with history of hypersensitivity to any vaccine and should be avoided in pregnancy.

Varicella Vaccine

Vaccination is necessary in following high-risk groups:

- Chronic cardiac or lung disease
- Asymptomatic HIV infection with CD4 >15 %
- Leukemia in remission and off chemotherapy >3 months
- Anticipated prolonged immunosuppression
- Prolonged aspirin therapy

- **Storage:** Freeze dried, lyophilized, 2-8^{0}C, use within 30 minutes of reconstitution.

- When used for post-exposure prophylaxis (within 72 hrs of contact), the protective efficacy of one dose of the vaccine is only 70%.

Rabies

Wound management:

Includes immediate irrigation with running water for 10 minutes, thorough cleaning with soap and water and application of povidone-iodine, 70% alcohol or tincture iodine.

Post exposure prophylaxis		
Category	**Type of Exposure**	**PEP Recommended**
I	Animal touch or lick on intact skin	Wound management
II	Minor scratches or abrasions without bleeding, licks on broken skin or nibbling of uncovered skin	Wound management and rabies vaccine
III	Single or multiple transdermal bites, abrasions that bleed, scratches or contamination of mucous membrane with saliva/ licks and any bite from a wild animal	Wound management and rabies vaccine + Rabies Immunoglobulin

Rabies immunoglobulin:

- Provides passive immunity by neutralizing rabies viruses.
- Its dose is 20 U/kg for human immunoglobulin and 40 U/kg for equine immunoglobulin or equine anti-rabies serum.
- This is infiltrated in and around the wound as soon as possible, not later than day 7 of injury.

Vaccination		
Essen or WHO	IM doses	Days 0, 3, 7, 14, 28
Updated Thai red cross	Two intradermal doses	Each on day 0, 3, 7 and 30

PEP For Previously Vaccinated Individuals:

- Local wound care
- No RIG needed
- 2 doses on Day 0 and Day 3.

PEP For Unvaccinated Individuals:

- Local wound care
- Three IM doses on days 0, 7 and 28

Yellow fever vaccine

- Live attenuated
- **Dose, route:** 0.5 mL, subcutaneous
- **Site**: Anterolateral thigh or upper arm
- Single dose ≥ 10 days before travel to endemic areas; revaccinate after 10 years.
- **Storage:** 2-8^{0}C; use within 30 minutes of reconstitution

Adverse events following immunization (AEFI)

Vaccine induced: Event caused or precipitated by an active component of vaccine, e.g.,

- Anaphylaxis after measles vaccine
- Vaccine associated paralytic poliomyelitis
- BCG related adenitis
- Pertussis encephalopathy

Components of vaccines implicated in mediating allergic reactions	
Egg protein	Yellow fever, measles, MMR, rabies PCEV, influenza killed injectable and live attenuated vaccines
Gelatin	Influenza, measles, MMR, rabies, varicella, yellow fever and zoster vaccines
Latex	In the rubber of the vaccine vial stopper or syringe plunger
Casein	• DTaP vaccine
Saccharomyces cerevisiae	• Hepatitis B • HPV vaccines

Vaccine specific serious adverse events	
Vaccine	**Adverse event**
Oral polio	Paralytic polio
Measles	Thrombocytopenic purpura within 7-30 days
Measles, Mumps and/ or Rubella	Encephalopathy or encephalitis <15 days
Tetanus	Brachial neuritis <=28 days
Pertussis	Encephalopathy or Encephalitis <=7 days
Rotavirus	Intussusception <= 30 days
Rubella	Chronic Arthritis <6 weeks.

Vaccine Storage and Cold Chain

- Vaccines such as OPV, measles and BCG (especially after reconstitution), are sensitive to heat but can be frozen without harm.

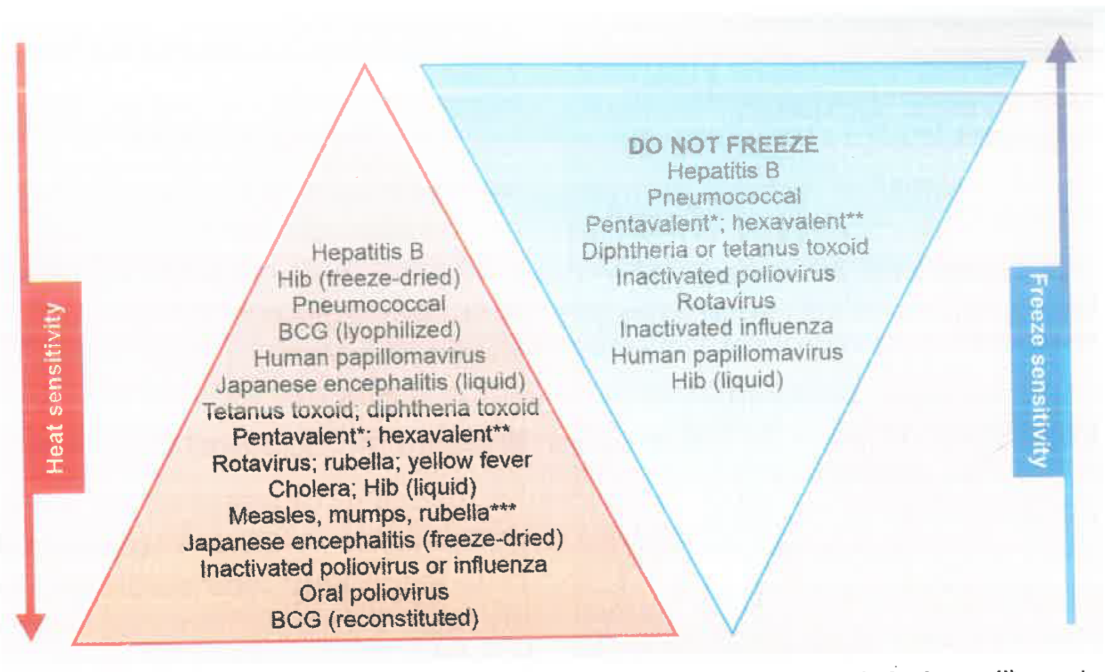
- In contrast, vaccines like hepatitis B, DTP, Pentavalent and tetanus toxoid are less sensitive to heat and damaged by freezing.

- In the refrigerator, OPV vials are stored in the freezer compartment (0 to -4°C).

- In the main compartment (4-10°C) BCG, measles and MMR are kept in the top rack (just below the freezer)

- **DTP, DT, TT, hepatitis A and typhoid** are stored in the **middle racks**

- **Hepatitis B, varicella and diluents** are stored in the **lower racks.**



Immunization limits in unimmunized child according to NIS	
• **BCG** • **Hep B** • **Rota virus** • **Fractionated IPV (fIPV)** • **Pneumococcal conjugated vaccine (PCV)**	Upto 1 yr of age
• **OPV** • **Measles/MR** • **Vit A**	Upto 5 yr of age
HiB	Upto 6 yrs of age
DPT	Upto 7 yrs
Japanese encephalitis (JE)	Upto 15 yrs of age